

利用如下公式：

$$\begin{aligned}\operatorname{sn}(u+v, m) &= \frac{\operatorname{sn}(u, m) \operatorname{cn}(v, m) \operatorname{dn}(v, m) + \operatorname{sn}(v, m) \operatorname{cn}(u, m) \operatorname{dn}(u, m)}{1 - m \operatorname{sn}^2(u, m) \operatorname{sn}^2(v, m)} \\ \operatorname{cn}(u+v, m) &= \frac{\operatorname{cn}(u, m) \operatorname{cn}(v, m) - \operatorname{sn}(u, m) \operatorname{dn}(u, m) \operatorname{sn}(v, m) \operatorname{dn}(v, m)}{1 - m \operatorname{sn}^2(u, m) \operatorname{sn}^2(v, m)} \\ \operatorname{dn}(u+v, m) &= \frac{\operatorname{dn}(u, m) \operatorname{dn}(v, m) - m \operatorname{sn}(u, m) \operatorname{cn}(u, m) \operatorname{sn}(v, m) \operatorname{cn}(v, m)}{1 - m \operatorname{sn}^2(u, m) \operatorname{sn}^2(v, m)}\end{aligned}$$

不难算出相应的二倍角公式：

$$\begin{aligned}\operatorname{sn}(2u|m) &\rightarrow -\frac{2\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)}{m\operatorname{sn}(u|m)^4 - 1} \\ \operatorname{cn}(2u|m) &\rightarrow \frac{-m\operatorname{sn}(u|m)^4 + 2\operatorname{sn}(u|m)^2 - 1}{m\operatorname{sn}(u|m)^4 - 1} \\ \operatorname{dn}(2u|m) &\rightarrow \frac{-m\operatorname{sn}(u|m)^4 + 2m\operatorname{sn}(u|m)^2 - 1}{m\operatorname{sn}(u|m)^4 - 1}\end{aligned}$$

三倍角公式：

$$\begin{aligned}\operatorname{sn}(3u|m) &\rightarrow \frac{m^2\operatorname{sn}(u|m)^9 - 6m\operatorname{sn}(u|m)^5 + 4m\operatorname{sn}(u|m)^3 + 4\operatorname{sn}(u|m)^3 - 3\operatorname{sn}(u|m)}{3m^2\operatorname{sn}(u|m)^8 - 4m^2\operatorname{sn}(u|m)^6 - 4m\operatorname{sn}(u|m)^6 + 6m\operatorname{sn}(u|m)^4 - 1} \\ \operatorname{cn}(3u|m) &\rightarrow \frac{-m^2\operatorname{cn}(u|m)\operatorname{sn}(u|m)^8 + 4m^2\operatorname{cn}(u|m)\operatorname{sn}(u|m)^6 - 6m\operatorname{cn}(u|m)\operatorname{sn}(u|m)^4 + 4\operatorname{cn}(u|m)\operatorname{sn}(u|m)^2 - \operatorname{cn}(u|m)}{3m^2\operatorname{sn}(u|m)^8 - 4m^2\operatorname{sn}(u|m)^6 - 4m\operatorname{sn}(u|m)^6 + 6m\operatorname{sn}(u|m)^4 - 1} \\ \operatorname{dn}(3u|m) &\rightarrow \frac{-m^2\operatorname{dn}(u|m)\operatorname{sn}(u|m)^8 + 4m\operatorname{dn}(u|m)\operatorname{sn}(u|m)^6 - 6m\operatorname{dn}(u|m)\operatorname{sn}(u|m)^4 + 4m\operatorname{dn}(u|m)\operatorname{sn}(u|m)^2 - \operatorname{dn}(u|m)}{3m^2\operatorname{sn}(u|m)^8 - 4m^2\operatorname{sn}(u|m)^6 - 4m\operatorname{sn}(u|m)^6 + 6m\operatorname{sn}(u|m)^4 - 1}\end{aligned}$$

四倍角公式：

$$\begin{aligned}\operatorname{sn}(4u|m) &\rightarrow \frac{-4m^3\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)^{13} + 8m^3\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)^{11} + 8m^2\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)^{11} - 20m^2\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)^9 + 20m\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)^5 - 8m\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)^3 - 8\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)^3 + 4\operatorname{cn}(u|m)\operatorname{dn}(u|m)\operatorname{sn}(u|m)}{3m^4\operatorname{sn}(u|m)^{16} - 16m^4\operatorname{sn}(u|m)^{14} + 16m^4\operatorname{sn}(u|m)^{12} - 4m^3\operatorname{sn}(u|m)^{14} + 40m^3\operatorname{sn}(u|m)^{12} - 20m^2\operatorname{sn}(u|m)^{14} + 20m^2\operatorname{sn}(u|m)^{12} - 4m\operatorname{sn}(u|m)^{14} + 40m\operatorname{sn}(u|m)^{12} - 20\operatorname{sn}(u|m)^{14} + 20\operatorname{sn}(u|m)^{12} - 4\operatorname{sn}(u|m)^{14} + 4\operatorname{sn}(u|m)^{12} - 1} \\ \operatorname{cn}(4u|m) &\rightarrow \frac{3m^4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{16} - 16m^4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{14} + 16m^4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} - 4m^3\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{14} + 40m^3\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} - 20m^2\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{14} + 20m^2\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} - 4m\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{14} + 40m\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} - 20\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{14} + 20\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} - 4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{14} + 4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} - 1}{3m^4\operatorname{sn}(u|m)^{16} - 16m^4\operatorname{sn}(u|m)^{14} + 16m^4\operatorname{sn}(u|m)^{12} - 4m^3\operatorname{sn}(u|m)^{14} + 40m^3\operatorname{sn}(u|m)^{12} - 20m^2\operatorname{sn}(u|m)^{14} + 20m^2\operatorname{sn}(u|m)^{12} - 4m\operatorname{sn}(u|m)^{14} + 40m\operatorname{sn}(u|m)^{12} - 20\operatorname{sn}(u|m)^{14} + 20\operatorname{sn}(u|m)^{12} - 4\operatorname{sn}(u|m)^{14} + 4\operatorname{sn}(u|m)^{12} - 1} \\ \operatorname{dn}(4u|m) &\rightarrow \frac{-m^5\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{18} + 4m^5\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{16} - 28m^4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} + 16m^4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{10} + 5m^4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{10} - 28m^3\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} + 16m^3\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{10} - 20m^2\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} + 20m^2\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{10} - 4m\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} + 40m\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{10} - 20\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} + 20\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{10} - 4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{12} + 4\operatorname{cn}(u|m)^2\operatorname{sn}(u|m)^{10} - 1}{3m^4\operatorname{sn}(u|m)^{16} - 16m^4\operatorname{sn}(u|m)^{14} + 16m^4\operatorname{sn}(u|m)^{12} - 4m^3\operatorname{sn}(u|m)^{14} + 40m^3\operatorname{sn}(u|m)^{12} - 20m^2\operatorname{sn}(u|m)^{14} + 20m^2\operatorname{sn}(u|m)^{12} - 4m\operatorname{sn}(u|m)^{14} + 40m\operatorname{sn}(u|m)^{12} - 20\operatorname{sn}(u|m)^{14} + 20\operatorname{sn}(u|m)^{12} - 4\operatorname{sn}(u|m)^{14} + 4\operatorname{sn}(u|m)^{12} - 1}\end{aligned}$$